Case Study

Shri Sant Gajanan Maharaj College of Engineering, Shegaon
Department of Electrical (Electronics & Power) Engineering
Course Title & Course Code: Power System Protection (8EP02)
Class: Final Year Electrical Engineering
Semester: VIII
Name of the Course Teacher: Prof. P. R. Bharambe
Title of the Innovative Practice: Use of YouTube Video Lectures for Conceptual Learning in Power System Protection

Activity Link: https://www.youtube.com/watch?v=ODHeHfOzS9c

Objectives/Goals of the Practice

The primary objective of this innovative teaching approach is to simplify complex relay protection mechanisms and fault analysis concepts using YouTube video lectures. The use of multimedia content is aimed at improving students' conceptual clarity, flexibility in learning, and exam preparation in the subject of Power System Protection.

Specific Objectives:

- To allow students to learn at their own pace through accessible, topic-focused video content.
- To enhance classroom learning through flipped classroom techniques and visual demonstrations.
- To visually explain relay characteristics, fault analysis, and system behavior using animations and simulation recordings.
- To provide revision material before practicals, tests, and final exams.
- To extend the resource base to learners from other institutions via open YouTube access.

Use of Appropriate Methods

To meet the above objectives, the following strategies were adopted:

- **Creation of Thematic Video Lectures:**
  High-quality video lectures were developed on key topics such as:
  - Protection of transmission lines
  - Working principles of electromagnetic relays
  - Differential protection of transformers
  - Distance protection, overcurrent protection
  - Circuit breaker operation and arc extinction
  - Protection coordination techniques
- **Video Structuring and Playlist Management:**
  All videos were organized into topic-wise playlists for easy navigation and sequential learning.
- **Integration with Teaching-Learning Process:**
  Students were encouraged to watch relevant videos before class. Classroom sessions then focused on interactive discussion, numerical solving, and clarification of advanced queries.

- **Feedback and Interaction:**
  Students were invited to ask doubts via the comment section or class WhatsApp group. Faculty responded promptly to queries to ensure continuity of learning.

Significance of Results

- **Flexible Learning:**
  Students could revisit complex topics multiple times for better clarity.
- **Boosted Exam Preparation:**
  Videos served as a go-to tool for quick revisions before university and internal exams.
- **Increased Reach:**
  Learners from nearby colleges also accessed and appreciated the content.
- **Improved Visual Understanding:**
  Visual demonstration of relay characteristics and protective schemes helped students grasp otherwise abstract concepts with greater ease.

Effective Presentation

- Videos were presented using clear audio narration, digital writing pads, and embedded diagrams/animations.
- Titles and thumbnails were designed to help students identify topics quickly.
- Videos were regularly reviewed for content quality, based on feedback and analytics (views, comments, average watch time).
- Video links were circulated through classroom groups and Learning Management Systems (LMS) for easy access.

Photos of the activity

POs Mapped: PO1, PO2, PO4, PO12
POs Mapped:PSO1

Reflective Critique

Faculty noted that the availability of video resources encouraged independent learning and better in-class participation. Many students who previously found relay and protection concepts difficult expressed improved confidence after using the video resources.

Peer Suggestions:

- Embed MCQs or quizzes at the end of videos for self-assessment.
- Add simulation-based examples for distance protection and differential protection.

Student Feedback:

- Students found video explanations to be simple and easy to follow.
- Several appreciated the pause-and-play facility, which helped them review tricky portions multiple times.
- Students found animation-based examples more helpful than textbook diagrams.

Evidences of Success

- More than **65% of students** accessed the YouTube video lectures during the semester.
- There was a noticeable improvement in test and oral exam performance for topics supported by videos.
- Students requested similar video support for protection system simulations using MATLAB or MiPower.

Challenges Faced During Implementation

- Some students from rural areas faced difficulty streaming videos due to weak internet connectivity.
- Creating detailed and accurate animations for protection system behavior was time-consuming.
- Maintaining high-quality audio and visual content required additional effort and equipment.
- A few students preferred live explanation and found asynchronous learning less engaging.